

Summer Internship Report

On

Artificial Intelligence Internship: Movie Recommendation System & Sentiment Analysis Dashboard

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By

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Internship Completion Certificate



CANDIDATE'S DECLARATION

I, Naitik Kumar, do hereby declare that the internship report titled “Artificial Intelligence Internship: Movie Recommendation System & Sentiment Analysis Dashboard” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

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Signature

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1. Introduction to the Internship Program

This report presents the work carried out during a one-month AICTE & ICAC approved Artificial Intelligence internship undertaken from 22/06/2026 to 22/07/2026 under the National Internship Portal, in association with Codec Technologies Pvt. Ltd. and Google for Education. The internship focused on practical, project-based learning in the field of Artificial Intelligence and Machine Learning, with an emphasis on building real-world applications using Python and modern data science libraries.

Over the course of the internship, two independent projects were undertaken: a collaborative-filtering based Movie Recommendation System, and an AI-powered Sentiment Analysis Dashboard. Both projects were designed, developed, tested, and deployed as interactive web applications using the Streamlit framework, giving hands-on exposure to the complete lifecycle of an AI/ML application — from data preprocessing and model building to deployment and user interaction.

2. Company Profile: Codec Technologies Pvt. Ltd.

Codec Technologies Pvt. Ltd. is an organisation engaged in delivering industry-oriented technical training and internship programs, in partnership with the National Internship Portal (Ministry of Education) and Google for Education. The internship program is structured to bridge the gap between academic learning and industry expectations by exposing students to real-world problem statements in emerging technology domains such as Artificial Intelligence, Machine Learning, and Data Science.

3. Project 1: Movie Recommendation System

3.1 Overview

The first project is a sleek, interactive movie recommendation system built using collaborative filtering on the MovieLens dataset. The application allows a user to select their profile and instantly receive a ranked list of movies they are likely to enjoy, based on the preferences of similar users.

3.2 Dataset

The MovieLens Small Dataset was used, consisting of 100,836 ratings across 9,742 movies contributed by 610 users.

3.3 Methodology

- Load and merge the ratings and movies datasets.

- Construct a User-Item ratings matrix of dimension $610 \times 9,719$.
- Compute Cosine Similarity between users to identify users with similar taste.
- Identify the top similar users for a given target user.
- Generate recommendations of unseen movies using similarity-weighted ratings.

3.4 Tech Stack

- Frontend & Deployment: Streamlit
- Language: Python
- Data Processing: Pandas, NumPy
- Machine Learning: Scikit-learn (Cosine Similarity)

3.5 Results

The model achieved an RMSE of 0.9754 evaluated on 500 test samples, indicating reasonably accurate rating predictions for the collaborative filtering approach. Sample recommendations generated for a test user included titles such as Terminator 2: Judgment Day (1991), Aliens (1986), The Sixth Sense (1999), The Hunt for Red October (1990), and The Godfather (1972).

3.6 Deployment

The application was deployed as a live, publicly accessible web app using Streamlit Community Cloud, allowing users to interact with the recommendation engine directly through a browser.

GitHub Repository: https://github.com/naitik2424/movie_recommendation-system

4. Project 2: AI Sentiment Analysis Dashboard

4.1 Overview

The second project is a versatile sentiment analysis dashboard that enables users to analyse the sentiment of text in real time, or to upload bulk CSV files of reviews/comments for automated, large-scale sentiment processing and visualization.

4.2 Key Features

- Dual-Model Support: toggle between a fast, rule-based lexicon model (NLTK VADER) and a state-of-the-art transformer model (Hugging Face DistilBERT).
- Real-Time Analysis: instantly analyse the sentiment intensity of single sentences or paragraphs.
- Bulk CSV Uploads: analyse thousands of reviews at once, with automatic visualization of sentiment distributions using Plotly pie charts and histograms.
- Data Export: download the fully processed, labelled dataset directly from the web interface.

4.3 Tech Stack

- Frontend & App Framework: Streamlit
- Language: Python
- Data Processing: Pandas
- Visualization: Plotly
- Natural Language Processing: NLTK, Hugging Face Transformers

4.4 Deployment

The dashboard was deployed on Streamlit Community Cloud, making it accessible for real-time and bulk sentiment analysis directly from a browser.

GitHub Repository: <https://github.com/naitik2424/sentiment-analysis-system>

5. Tools and Technologies Used

- Programming Language: Python
- Libraries: Pandas, NumPy, Scikit-learn, NLTK, Hugging Face Transformers, Plotly
- Framework: Streamlit
- Version Control & Hosting: Git, GitHub
- Deployment Platform: Streamlit Community Cloud

6. Learning Outcomes

- Gained practical experience in building end-to-end Machine Learning applications, from raw data to a deployed product.
- Learned collaborative filtering techniques and similarity-based recommendation algorithms.
- Learned to apply both classical (lexicon-based) and modern (transformer-based) NLP techniques for sentiment analysis.
- Gained hands-on experience with the Streamlit framework for rapidly building and deploying interactive data applications.
- Improved skills in data visualization using Plotly for presenting analytical results clearly.
- Developed familiarity with the complete deployment workflow using Git, GitHub, and Streamlit Community Cloud.

7. Conclusion

The one-month AI internship with Codec Technologies Pvt. Ltd. provided a valuable opportunity to apply theoretical concepts of Machine Learning and Natural Language Processing to practical, deployable projects. Through the development of the Movie Recommendation System and the AI Sentiment Analysis Dashboard, I gained hands-on experience across the full AI application lifecycle — including data preprocessing, model building, evaluation, visualization, and deployment. This internship has strengthened my technical foundation in AI/ML and has been instrumental in preparing me for future academic and industry-oriented work in this domain.

8. References

1. MovieLens Dataset, GroupLens Research, University of Minnesota.
2. Streamlit Documentation — <https://docs.streamlit.io>
3. Scikit-learn Documentation — <https://scikit-learn.org>
4. NLTK VADER Sentiment Analysis — <https://www.nltk.org>
5. Hugging Face Transformers Documentation — <https://huggingface.co/docs/transformers>

6. Movie Recommendation System Repository — https://github.com/naitik2424/movie_recommendation-system
7. Sentiment Analysis System Repository — <https://github.com/naitik2424/sentiment-analysis-system>